MIL-N-48153 (PA)

22 January 1974

MILITARY SPECIFICATION

NOSE LUCKY AND INSULATOR ASSEMBLY FOR CARTRIDGE 105MM, HEAT-T, M456A1E1 PARTS FOR, ASSEMBLY AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for assembling and packing for one type of Nose Lucky & Insulator Assembly for use with Cartridge, 105MM, HEAT-T, M456A1E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-P-12815 - Power Source, Piezoid (Lucky)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts

FSC: 1315

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Sampling by Attributes.

DRAWINGS

ARMY

9256503 - Nose Lucky & Insulator Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assemblies.-The assemblies shall comply with all requirements specified on Dwg. 9256503 and with all requirements specified in applicable specifications.

3.3 Lucky.-All luckies must conform to the requirements of Dwg. 9271693 and MIL-P-12815.

3.4 Activity.-The nose lucky assembly shall indicate activity after oven curing.

3.5 Insulator voltage breakdown.-The insulator shall withstand an application of 2000 volts min. DC for a minimum duration of 2 seconds without any visible disruptive discharge.

3.6 Adhesion.-Adhesive bonded parts shall not become loose or unbonded after oven curing.

3.7 Workmanship.-The assembly and all components shall be of the proper finish, dry and free of burrs, cracks, chips, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

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3.8 First article inspection.-This specification provides for submission of a first article sample in 4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will

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not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the initiation of regular production, the contractor shall submit a first article sample to the Government facility designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) assemblies and twenty-five(25) complete sets of parts which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment, which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply so as to significantly, affect product uniformity as determined by the Government).

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of applicable requirements for the inspections performed in accordance with 4.3.2 and 4.3.3 the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of assembly or component in the sample to comply with any or all of the stated requirements.

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4.3 Inspection Provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedures which is being applied to determine conformance with requirements. Lot numbering as required shall be in accordance with MIL-STD-1168.

4.3.1.1 Curing batch (see 4.3.3.3).-A batch shall be defined as the number of assemblies cured under the same conditions, in the same oven, at one time.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph heading. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to the individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of

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the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Plug (see dwg. 9271692)

Categories	Defects	Method of Inspection (See 6.2)
Critical:	None defined.	
Major:	AQL 0.65 percent	
101.	Diameter of prong incorrect	Gage
102.	Thickness of flange incorrect	Gage
103.	Flatness of face incorrect	Gage
Minor:	AQL 2.50 percent	
201.	Total length of plug incorrect	Gage
202.	Concentricity of flange diameter with prong incorrect	Gage
203.	Perpendicularity of face with prong incorrect	Gage
204.	Radius missing from end of prong	Visual
205.	Diameter of flange incorrect	Gage
206.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.2 Plunger (see dwg. 9256501)

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 1.00 percent	
101.	Outside diameter at datum incorrect	Gage

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102.	Runout at datum diameter incorrect	Gage
103.	Flange thickness incorrect	Gage
104.	Wall angle incorrect	Gage
105.	Overall height of plunger incorrect	Gage

Minor: AQL 2.50 Percent

201.	Thickness of base incorrect	Gage
202.	Side wall thickness incorrect	Gage
203.	Flange diameter incorrect	Gage
204.	Cup radius incorrect	Gage
205.	Flatness incorrect	Gage
206.	Evidence of poor workmanship (See 3.7)	Visual

4.3.2.3 Nose Lucky Assembly (See dwg. 9256502)

Categories	Defects	Method of Inspection

Critical: None defined.

Major: AQL 0.65 percent

101.	True position incorrect	Gage
102.	Cement on visible area of lucky	Visual
103.	Distance from plunger to edge of crystal under min	Gage

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104. Positive surface of lucky not assembled against plug.....Visual

Minor: AQL 0.65 percent
201. Evidence of poor workmanship (see 3.7).....Visual

4.3.2.4 Washer, insulator (see dwg. 9271035)

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
	101. Thickness of insulator washer incorrect.....	Gage
	102. Outside diameter incorrect.....	Gage
	103. True position of outside diameter incorrect.....	Gage
Minor:	AQL 1.50 Percent	
	201. Slot over max.	Gage
	202. Presence of Flash.....	Visual
	203. Inside diameter incorrect.....	Gage
	204. Evidence of poor workmanship (See 3.7).....	Visual

4.3.2.5 Insulator (see dwg. 8861462)
(Inspections to be performed while item is in restrained position).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
	101. Outside diameter incorrect.....	Gage
	102. Thickness of insulator incorrect..	Gage
	103. Material defective or damaged.....	Visual
Minor:	AQL 1.00 percent	
	201. Presence of flash.....	Visual
	202. Evidence of poor workmanship (see 3.7).....	Visual

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4.3.2.6 Nose Lucky and Insulator Assembly (See dwg. 9256503)

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Any component missing, damaged or improperly assembled	Visual
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.7)	Visual

4.3.3 Testing

4.3.3.1 Activity test (see 3.4).-Major defect - This test shall be conducted 100 percent. Any assembly which fails to comply with the applicable requirement shall be classified defective and be removed from the lot. This test shall be performed using contractor designed equipment in accordance with 4.4.1.

4.3.3.2 Insulator voltage breakdown (see 3.5).-Major defect - - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level II, AQL 0.40%. This test shall be performed using contractor designed equipment in accordance with 4.4.2.

4.3.3.3 Adhesion (see 3.6).-Major defect - Samples selected from each curing batch (see 4.3.1.1) shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level I, AQL 0.40%. Testing shall be performed using contractor designed equipment in accordance with 4.4.3.

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" heading in 4.3.2 and in 4.3.3 (See 6.2).

4.4 Test Methods and Procedures

4.4.1 Activity Test.-Using equipment specified in 4.3.4, the output activity of the nose lucky assembly must comply with the applicable requirement. An impact load shall be applied to the face of the lucky. The test method used shall not damage the nose lucky assembly.

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4.4.2 Insulator voltage breakdown.-The insulator shall be subjected to 2000 volts DC minimum, applied to the inside and outside faces. The test shall be performed in accordance with Method 301 of MIL-STD-202, with visual observation for voltage breakdown (see 6.4). The equipment for this test shall be in accordance with 4.3.4.

4.4.3 Adhesion.-The nose lucky assembly shall be mounted firmly by the plug component in a suitable fixture and a 14 to 15 lb shear load applied to the plunger immediately adjacent to the plunger flange. The mounting shall provide support immediately adjacent to the plug flange. The equipment for this test shall be in accordande with 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-Packing and marking shall be in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Itspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3, should he elect to do so.

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6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (See 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification, paragraph and defect numbers from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will

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be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 It is noted that the size of inspection lots of component assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.4 Voltage Breakdown.-Voltage breakdown is a visible disruptive discharge as evidenced by flashover (surface discharge) or sparkover (air discharge) on the insulator surface (See 4.4.2).

6.5 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Submission of Test Data and Ammunition Data Cards.-In addition to the normal distribution of records, when the assembly is procured by the Army Armament Command, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A and
SARPA-AD-E-A
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1315-A-016

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